Cracking the Black Box

Portfolio Replica Strategy — BC3 final deliverable

PoliMI Fintech — Business Case 3

Problem and data

Goal. Replicate a hidden "Monster Index" using only liquid Bloomberg futures, subject to a regulatory VaR cap.

$$r_t^{\text{target}} = 0.50\, r_t^{\text{HFRX}} + 0.25\, r_t^{\text{MSCI World}} + 0.25\, r_t^{\text{Global Bond}}$$

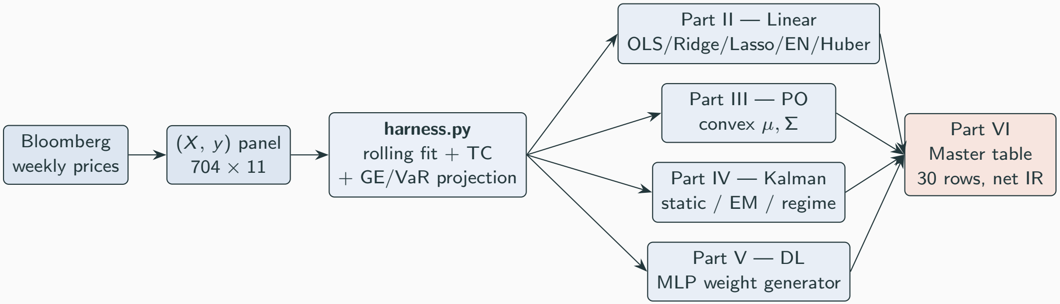
Universe (11 futures).

- Rates: RX1, TY1, DU1, TU2
- Equity: ES1, NQ1, VG1, TP1, LLL1
- Commodity: GC1, CO1

Constraints.

- Sample: weekly, Oct 2007 – Apr 2021 (704 obs).
- Gaussian VaR cap (1%, 1-month) $\leq 8\%$.
- Gross-exposure cap: $\sum_j |w_j| \leq L$.
- Transaction cost: $\tau = 5$ bps on L1 turnover.

Pipeline overview — one harness, four model families



Shared contract. Every model returns a `ReplicaResult`: `weights_history`, `replica_returns`, `target_returns`, `metrics`. Part VI loads them all into one ranked leaderboard.

Backtest methodology — Part I

Rolling walk-forward. At every rebalance week t:

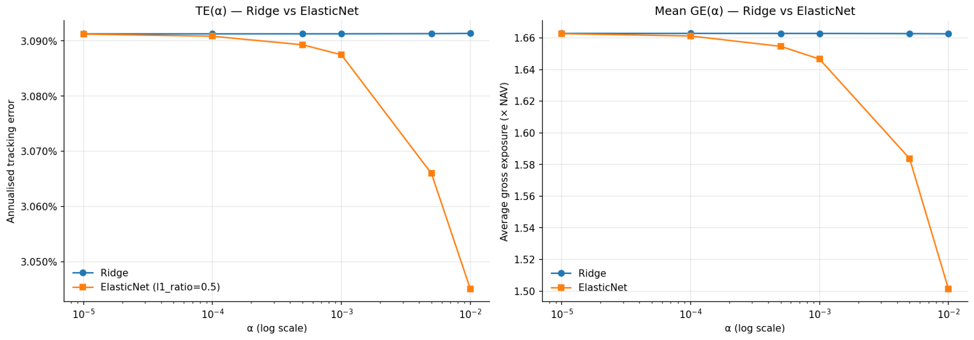
1. **Fit** on $X[t{-}104:t]$, $y[t{-}104:t]$ (no look-ahead — $X[t]$, $y[t]$ never seen).
2. **Project** onto the feasible set: gross cap $\sum_j |w_j| \le L$, then iterative shrinkage until $\mathrm{VaR}(w) \le 8\%$.
3. **Trade** $\Delta w_t = w^{\text{new}} - w^{\text{old}}$; **cost** $c_t = \tau \cdot \sum_j |\Delta w_{t,j}|$.
4. **Realise** $r_t^{\text{rep}} = w_t^\top r_t$; on non-rebalance weeks carry w_{t-1} forward.

Contract metrics.

$$\mathrm{TE} = \sqrt{52\,\mathrm{Var}(r_t^{\text{rep}} - y_t)}, \quad \mathrm{IR} = \frac{52\,\mathbb{E}[r_t^{\text{rep}} - y_t]}{\mathrm{TE}}, \quad \mathrm{net\_IR} = \frac{52\,\mathbb{E}[r_t^{\text{rep}} - c_t - y_t]}{\mathrm{net\_TE}}$$

VaR fallback. Jarque-Bera on the target rejects normality ($p \approx 0$); Gaussian VaR (4.14%) understates the historical 1%-quantile (5.96%) — the harness can fall back to a historical VaR projector.

Part II — classical linear benchmark



Ridge vs. ElasticNet across matched α grid: TE flat in α; ElasticNet's average gross exposure collapses once L1 starts zeroing coefficients at $\alpha \geq 10^{-3}$.

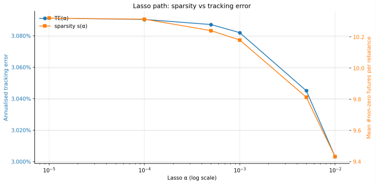
Sweep. {OLS, Ridge, Lasso, EN, Huber} $\times$ {all-11, top-5} — single harness: 104-week window, $\Delta t = 4$, scale-only, GE cap 2, VaR cap 20%.

Reading.

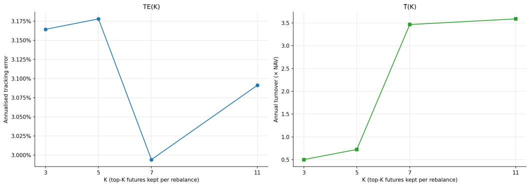
- Ridge / EN collapse to OLS at $\alpha < 10^{-3}$.
- Top-5 pruning saves $\sim 4\times$ on average GE for a small TE premium.

Part II — why these modelling choices?

1. **Lasso & sparsity.** L_1 drives weights to zero, cutting active assets & turnover. *But* excessive sparsity underfits — too few futures cannot track a multi-asset target.
2. **Top-K selection.** Removing low-corr futures reduces exposure & stabilises turnover. *But* too small a K discards useful signal with the noise.
3. **Huber robust loss.** Down-weighting outliers limits crisis-month damage. *But* it is not necessarily optimal over the full sample — the δ that protects in 2008/2020 can lag in calm regimes.
4. **Ridge / ElasticNet.** Moderate L_2 shrinkage is strictly more stable than unregularised OLS. *But* the gain vanishes at $\alpha < 10^{-3}$.



Lasso sparsity vs. tracking error: zeroing too many weights inflates TE.



Top-K: TE rises steeply below $K = 5$; turnover drops monotonically.

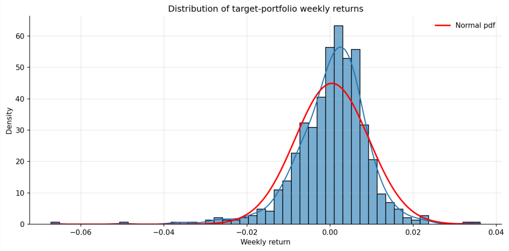
Part III — portfolio constraints (post-hoc projection)

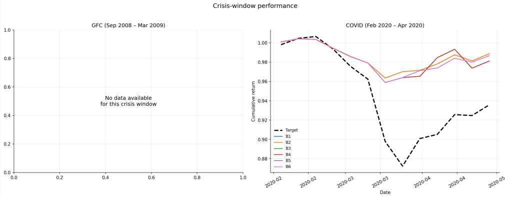
Why projection matters. Gaussian VaR is rejected by Jarque-Bera; the historical 1%-quantile is fatter than $\mathcal{N}$. The harness projects w onto a *historical* VaR feasible set.

Six base configurations (B1–B6):

$$\Delta t \in \{1, 4, 13\}\ \text{wk} \ \times \ L \in \{1.0,\ 2.0\}$$

Crisis-window evaluation. 2008-09 $\rightarrow$ 2009-03 (GFC) and 2020-02 $\rightarrow$ 2020-04 (COVID). The COVID drawdown is the binding test: every config beats the target on max-drawdown, but only the quarterly cadences keep TE bounded.





Part III — predict-then-optimize (convex layer)

Two-stage rebalance.

1. **Alpha.** Fit a classical linear model on $(X_{\text{tr}}, y_{\text{tr}})$; take $\mu = \texttt{coef\_}$.
2. **Risk.** Ledoit–Wolf shrunken sample covariance Σ (annualised).
3. **Optimise** (CVXPY):

$$\max_{w} \;\; \mu^\top w \; - \; \tfrac{1}{2}\lambda \, w^\top \Sigma \, w \; - \; \tau \, \|w - w_{t-1}\|_1 \quad \text{s.t.} \quad \|w\|_1 \le L, \;\; |w_j| \le 0.5$$

Sanity check. With $\lambda = 2$, slack constraints, no LW shrinkage, the convex layer recovers OLS to within solver tolerance.

Findings.

- Longest rolling window (3y) and modest Ridge dominate net IR on the leaderboard.
- Huber is competitive only in the 2008/2020 windows.
- Lasso's sparsity buys lower turnover but loses on TE.

Part IV — Kalman filter / state-space formulation

$$x_t = A\,x_{t-1} + B\,u_t, \qquad u_t \sim \mathcal{N}(0, I_{11}) \quad \text{(state: portfolio weights, random walk)}$$
$$y_t = C_t\,x_t + D\,\varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, 1) \quad \text{(observation: target return)}$$

$A = I_{11}$, $B = \sigma_w \cdot I_{11}$, $C_t = r_t^\top$, $D = \hat{\sigma}_y$ from the training window, $x_t \in \mathbb{R}^{11}$ the latent weight vector, y_t the scalar target return. Ridge regression warm-start on the first 52 weeks; filter implemented from scratch.

Three variants:

- **Basic Kalman** — fixed process noise $\sigma_w = 0.001$, fixed R, with a VaR guardrail.
- **Tuned Kalman** — grid search over $(Q, R_{\text{scale}}, P_0)$ with a Universal Replication Score, plus Q-sensitivity analysis.
- **Double Adaptive Volatility** — both Q and R scale dynamically with rolling volatility.

Parameter Robustness. $\pm 20\%$ perturbation grid confirms structural soundness.

Look-ahead discipline. $\hat{x}_{t|t}$ uses y_t (unavailable at the open of week t). We shift weights by one week: held weight at t depends only on observations through $t-1$.

Part IV — Kalman constraints & optimisation score

1. Turnover Dampening

$$\hat{w}_t^{\text{damp}} = (1-\alpha)\,\hat{w}_{t-1}^{\text{final}} + \alpha\,\hat{x}_{t|t}$$

Exponential smoothing on raw Kalman updates, structurally slowing weight adjustments to prevent excessive trading and reduce transaction costs.

2. Gross Exposure Cap

$$\hat{w}_t^{\text{gross}} = \begin{cases} \hat{w}_t^{\text{damp}} \dfrac{GE_{\text{cap}}}{\sum_i |\hat{w}_{i,t}^{\text{damp}}|} & \sum_i |\hat{w}_{i,t}^{\text{damp}}| > GE_{\text{cap}} \\ \hat{w}_t^{\text{damp}} & \text{otherwise} \end{cases}$$

Proportional scale-down if the absolute sum of weights exceeds $GE_{\text{cap}} = 2.0$.

3. VaR Guardrail

$$\hat{w}_t^{\text{final}} = \begin{cases} \hat{w}_t^{\text{gross}} \dfrac{VaR_{\text{cap}}}{\widehat{VaR}} & \widehat{VaR} > 8\% \\ \hat{w}_t^{\text{gross}} & \text{otherwise} \end{cases}$$

where $\widehat{VaR} = VaR_{1\%,\,4W}(\hat{w}_t^{\text{gross}})$. Aggressively downweights if forward risk breaches 8%.

4. Replication Score

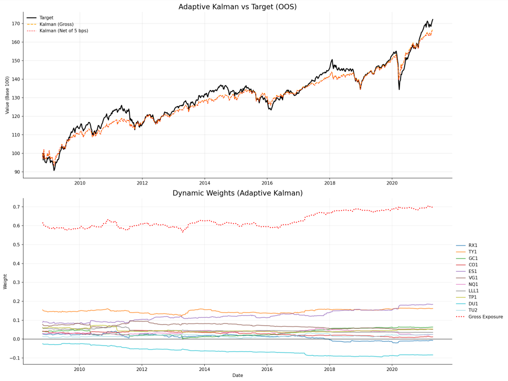
$$S = \rho(\mathbf{r}_R, \mathbf{r}_T) - 2\,TE_n - 0.5\,\overline{\Delta w} - 0.1\,MDD_n$$

TE_n = net TE, $\overline{\Delta w}$ = avg turnover, MDD_n = net max drawdown.

Part IV — Adaptive Kalman: results comparison

Metric	Target	Basic Kalman	Adaptive Kalman
Ann. Return	4.52%	4.06%	**4.20%**
Ann. Volatility	6.11%	5.64%	**5.55%**
Sharpe	0.74	0.72	**0.76**
Max Drawdown	13.35%	9.30%	**8.81%**
Tracking Error	N/A	3.11%	**3.04%**
Info. Ratio	N/A	−0.1491	**−0.1046**
Correlation	N/A	0.8633	**0.8684**
Avg Turnover	N/A	0.96%	**0.55%**
Avg Gross Exp.	N/A	0.8559	**0.6278**
Avg VaR	N/A	3.61%	**3.57%**
Cost Drag	N/A	0.02%	**0.01%**

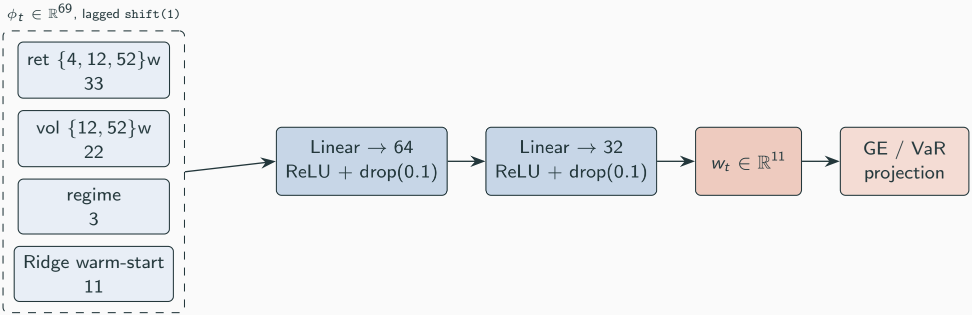
Adaptive improves return, TE, drawdown, turnover, and gross exposure.



Adaptive Kalman replica vs. target cumulative returns.

Part V — deep learning weight generator (architecture)

End-to-end. The network outputs the *weights*, not the returns: $w_t = f_\theta(\phi_t)$, $r_t^{\text{rep}} = w_t^\top r_t$.



Loss — forward-window MSE + turnover + budget:

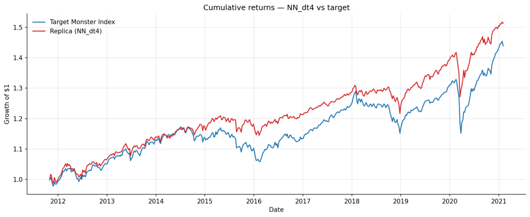
$$\mathcal{L}(\theta) = \underbrace{\mathrm{MSE}_{s\in[t,\,t+H-1]}\big(w_t^\top r_s - y_s\big)}_{H=12} + \lambda_1 \|\Delta w\|_1 + \lambda_2 \|\Delta w\|_2^2 + \gamma\,(1^\top w_t - 1)^2$$

Part VI — consolidated leaderboard (top of 30 rows)

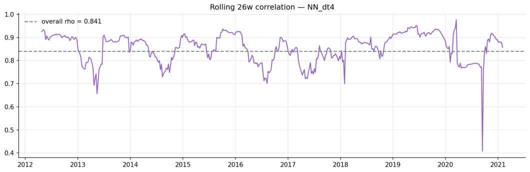
Model	net_IR	IR	ρ	TE	GE	$\bar{T}$
NN $\Delta t = 4$	**+0.170**	+0.177	0.841	2.96%	1.02	0.008
NN $\Delta t = 1$	+0.135	+0.143	0.843	2.93%	1.03	0.010
M3-PO | Huber $\delta = 1.35$ | all-11	+0.017	+0.032	0.856	2.93%	1.28	0.017
M3-PO | OLS | all-11	+0.017	+0.032	0.853	2.95%	1.38	0.018
M2 | Huber | all-11	-0.086	-0.022	0.844	3.03%	1.59	0.075
Kalman (basic, $\sigma_w = 10^{-3}$)	-0.141	-0.137	0.868	3.04%	0.59	0.005
M2 | Lasso / EN | top-5	-0.185	-0.173	0.828	3.18%	0.37	0.014
M3-Cstr | B1 (weekly / 100% cap)	-0.468	-0.461	0.765	3.91%	0.19	0.011

Headline. NN $\Delta t = 4$ tops the leaderboard on net IR with $\rho = 0.84$ at the lowest gross exposure and turnover of the positive-IR group. The gap to the next cluster (PO + Huber / OLS, $\rho \approx 0.85$) is narrow; M3-Cstr (post-hoc EN at 1y window) loses the most — the constraint sweep without the right model class cannot rescue tracking quality.

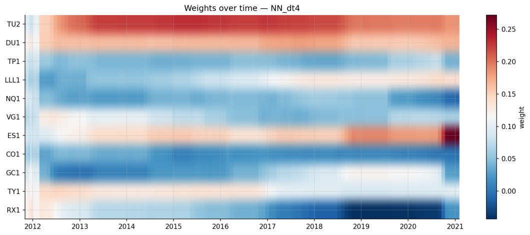
Best model spotlight — cumulative & weights



NN $\Delta t = 4$ vs target, OOS 2011-11 $\rightarrow$ 2021-02.



Rolling 26-week correlation; overall $\rho = 0.84$.



Asset weights over time (RdBu = long/short).

Reading the heatmap.

- Persistent long in rates (RX1, TY1) carries the bond leg.
- Equity loadings (ES1, NQ1, VG1) rotate with the regime.
- Sign-flips are mild — the network respects the turnover penalty.

Findings

- **Methodology before models.** Part I's harness fixes rolling-window discipline, cost accounting, and the VaR / GE projection *once*; every Part II–V model plugs into the same evaluation surface and lands in the same master table.
- **Cadence sets the cost/fit trade-off.** Weekly rebalancing maximises tracking but pays the heaviest 5 bps drag; quarterly minimises drawdown but loses precision. The B1–B6 sweep makes the trade-off auditable.
- **Long window beats reactive fitting.** 3-year rolling dominates 1-year on net IR across every linear model — weekly data is too noisy to adapt aggressively.
- **Kalman EM matches the static $\sigma_w = 10^{-3}$ grid point.** The regime-switching variant narrows TE in 2008 at the cost of turnover.
- **The MLP wins net IR.** `NN_dt4` delivers $\rho = 0.84$ at $\overline{\mathrm{GE}} \approx 1.0$ and $\bar{T} \approx 0.8\%$; without $\lambda_1 \|\Delta w\|_1$, NN variants over-trade and lose net IR.
- **Test vs. val gap is regime, not overfit.** `train` $\approx$ `val` $\ll$ `test` across λ_1 values — the test fold crosses COVID; the model is honest about this.